Conor Pedroche

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Education

University of California, Berkeley Expected May 2030

B.A. in Computer Science

Relevant Coursework: Introduction to Computer Science, Introduction to Programming with Python, Structure and Interpretation of Computer Programs, Data Structures, Intro to Computing with HTML, Fundamentals of Programming I

Experience

UCSB Summer Research Academies Jun 2024 – Jul 2024

Research Intern *Santa Barbara, CA*

- Collected and processed **1,802 Reddit discussion threads, 18,020 comments, and 101 news outlets** using the **Reddit API, Selenium, and Python** to prepare a dataset linking news sources, discussion threads, and audience comments for analysis.
- Applied **VADER sentiment analysis**, **correlation analysis**, and **multiple linear regression** to study relationships between media reputation, article tone, political topics, and audience responses, comparing article sentiment with reader reactions.
- Co-authored and presented a research paper at the **Capstone Seminar**, analyzing the relationship between media outlet reputation and its influence on political bias among Reddit audience responses and their interactions with each other.

Self-Employed Sep 2020 – Aug 2026

PC Builder / Technical Advisor *Santa Maria, CA*

- Designed, built, and optimized custom gaming PCs for **30+ customers**, selecting components based on budget, performance, and aesthetics, using PCPartPicker to research and compare CPUs, GPUs, PSUs, cooling systems, and other components.
- Diagnosed and resolved hardware and software failures, including POST errors, boot loops, overheating, memory instability, driver conflicts, system crashes, and BSODs through BIOS/UEFI configuration, component isolation, and stress testing.

Society of Hispanic Professional Engineers (SHPE) Aug 2025 – Jun 2026

President *Santa Maria, CA*

- Led monthly STEM meetings and **11 hands-on engineering projects** by creating technical presentations for members, organizing competitive activities, and guiding teams through project planning, design challenges, testing, and final builds.
- Recruited and retained **20+ new members**, expanding outreach to underrepresented students and strengthening their collaborative skills, technical and critical thinking, team design, and practical application of engineering concepts.
- Collaborated with Cal Poly students to organize STEM workshops and campus visits for local high school students.

Projects

Sourcecado | *Python*, *TypeScript*, *FastAPI*, *React*, *SQLite*, *Docker*, *MCP*

- Built a local-first AI desktop application with **FastAPI, WebSockets, React/TypeScript, Tauri, and SQLite**, utilizing **228 pytest and 35 Vitest** to cover migrations, retries, message attribution, and failure recovery for a Gmail connection feature.
- Implemented **fault-tolerant recovery** for interrupted AI agent workflows, restoring saved responses, resuming eligible runs under their original IDs, and preventing duplicate messages and external actions through idempotent processing.
- Centralized **approval validation** across chat and contact-management workflows, binding external actions to verified recipients and reviewed message content, validating enrichment requests, and maintaining authorization in API entry points.

Campus Event Finder | *Python*, *Flask*, *SQLite*, *HTML/CSS*, *Git*

- Built a **full-stack** web application for students to search and filter campus events by date, category, and location on campus.
- Designed a **Flask REST API** for event creation, retrieval, filtering, and deletion, using SQLite for persistent storage.
- Implemented user authentication and input validation to protect accounts and prevent invalid event submissions in the app.
- Added **pytest unit tests** and used Git/GitHub for version control, debugging, and feature development across the project.

Technical Skills

Languages: Python, C, TypeScript, SQL, HTML/CSS
Technologies/Frameworks: React, FastAPI, Flask, Tauri, SQLite, Selenium, VADER, REST APIs, WebSockets, MCP
Developer Tools: Vitest, pytest, Git, GitHub, Docker, VS Code

Honors & Awards

Regents' and Chancellor's Scholar | SkillsUSA Cybersecurity — State and Regional Gold Medalist